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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2017-2018

CE2005/CZ2005 – OPERATING SYSTEMS

Nov/Dec 2017 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 7 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions **DO NOT** carry equal marks.

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1. (a) Indicate whether the following statements are true or false. Justify your answers.

(i) Round-robin scheduling is also a form of priority-based scheduling in which each process has a fixed priority.

(2 marks)

(ii) The resource-allocation graph of Figure Q1 shows a deadlocked system.

(2 marks)

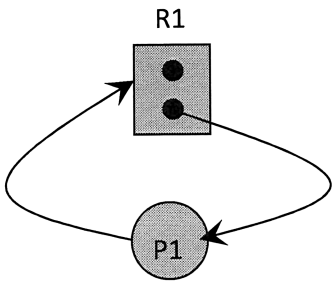


Figure Q1

(iii) Abort() system call is used by a child process to terminate its own execution.

(2 marks)

Note: Question No. 1 continues on Page 2

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(iv) Direct Memory Access (DMA) based I/O operations do not use interrupts.

(2 marks)

(v) Multiprogramming is not possible on a Central Processing Unit (CPU) with multiple cores.

(2 marks)

(vi) I/O protection is achieved by ensuring that all I/O instructions can only be executed in the kernel mode.

(2 marks)

(b) Consider the following set of processes to be scheduled.

Table Q1

Process	Arrival Time	Arrival Order	CPU Burst Length
P1	0	0	6
P2	0	1	5
P3	3	-	1
P4	7	-	2

Suppose we wish to assign priority values 1, 2, 3 and 4 to these processes such that a lower value indicates a higher priority, and assume we use preemptive priority-based scheduling. Answer the following questions, showing the set of processes in the ready queue at each time instant when the scheduler executes.

(i) Allocate priorities to processes in Table Q1 such that the schedule is identical to Shortest Job First (non-preemptive).

(3 marks)

(ii) What is the minimum quantum size under Round-Robin scheduling such that an identical schedule can be generated using preemptive priority-based scheduling for the processes in Table Q1? Also identify the resulting process priorities.

(4 marks)

Note: Question No. 1 continues on Page 3

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(c) Consider the following user-level solution to the critical section problem, where **flag** and **turn** are shared variables initialized to **false** and **0**, respectively. Does it satisfy mutual exclusion? Justify your answer.

Process P0

```
while(1){
   flag=true;
   while(turn==1);
   critical-section
   turn=1;
   remainder-section
}
```

Process P1

```
while(1){
   turn=0;
   while(flag and (turn==0));
   critical-section
   flag=false;
   remainder-section
}
```

(6 marks)

2. (a) Indicate whether the following statements are true or false. Justify your answers.

(i) A counting semaphore initialized to value 1 can be used to ensure mutually exclusive access to critical sections.

(2 marks)

(ii) Under non-preemptive scheduling, the scheduler always executes when a new process is inserted into the ready queue.

(2 marks)

(iii) When a process issues a system call, hardware uses trap to transfer control to the kernel.

(2 marks)

(iv) Process instructions are stored in the "Text" segment of process memory.

(2 marks)

(v) If the circular-wait condition is satisfied for a set of processes and resources, then deadlock has occurred.

(2 marks)

(vi) Inter-process communication (IPC) using shared memory is fast when compared to message passing.

(2 marks)

Note: Question No. 2 continues on Page 4

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(b) Consider the following snapshot of a system's state, with four processes (P1, P2, P3 and P4) and three resource types (A, B and C). The current Allocation, Need and Available matrices are as shown in Table Q2.

Table Q2

		Available
		A B C
		0 2 0
Process	**Allocation**	**Need**
	A B C	A B C
P1	0 1 1	1 1 0
P2	1 1 0	0 1 1
P3	1 1 1	2 0 0
P4	1 1 1	0 1 0

(i) What are the total number of instances of each resource type (A, B and C) in this system?

(3 marks)

(ii) Is the above system state safe? Justify your answer.

(4 marks)

(c) Consider the Reader-Writer synchronization problem in which several "Reader" processes read a shared data D and several "Writer" processes modify the same shared data D. **For correctness purposes, Writers must have exclusive access to D (one at a time), whereas multiple Readers can read D simultaneously**.

One solution to this problem is presented below in which the Readers have preference over Writers. That is, once a Reader has gained access to D, other Readers can continue to read D even if a Writer is waiting to get access to D. In this solution, **res** and **r_mutex** are binary semaphores and **readcount** is an integer variable. **res** and **r_mutex** are initialized to **1** and **readcount** is initialized to **0**.

Modify the below processes, using additional binary semaphores and variables if necessary, such that Writers are given preference over Readers. That is, once a Writer requests access to D, no Reader should be allowed to gain access to D until the Writer has completed modifying D. You can assume that processes blocked on a semaphore are woken up in First-Come First-Served order.

Note: Question No. 2 continues on Page 5

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Writer

```
while(1) {
   wait(res);
   Update D;
   signal(res);
}
```

Reader

```
while(1) {
   wait(r_mutex);
   readcount++;
   if(readcount==1) wait(res);
   signal(r_mutex);
   Read D;
   wait(r_mutex);
   readcount--;
   if(readcount==0) signal(res);
   signal(r_mutex);
}
```

(6 marks)

3. (a) Indicate whether the following statements about memory management are true or false. Justify your answers.

(i) Logical address and physical address will be identical if address binding is performed during compile time.

(2 marks)

(ii) Memory compaction is used to reduce internal fragmentation.

(2 marks)

(iii) Context switch contributes the most to the overhead of handling page faults.

(2 marks)

(iv) Using demand paging for virtual memory support, CPU utilization can be always improved by increasing the degree of multi-programming.

(2 marks)

(v) Fixed frame allocation policy implies that the page to be replaced must be chosen from the frames allocated to the same process.

(2 marks)

Note: Question No. 3 continues on Page 6

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(b) Consider a computer system with a 10-bit logical address. Assume paging is used for memory allocation and the page size is 256 bytes. There are 32 frames in physical memory.

(i) What is the size of physical memory?

(2 marks)

(ii) What are the logical and physical address formats?

(3 marks)

(iii) Given the page table shown in Figure Q3b, translate logical address 1001001010 to the corresponding physical address.

Page Number	Frame Number
0	12
1	5
2	16
3	7

Figure Q3b

(3 marks)

(c) Assume a computer system uses Translation Look-aside Buffer (TLB) with page table for address translation. Explain whether or not the following scenarios can happen:

- TLB hit with no page fault
- TLB hit with page fault
- TLB miss with no page fault
- TLB miss with page fault

(6 marks)

(d) Assume a computer system uses demand paging for virtual memory management. There are three page frames available which are initially empty. Figure Q3d shows the behavior of the Least Recently Used (LRU) page replacement algorithm on a page reference string.

F1	2	2	2	2	2	2	2	2
F2		3	3	3	5	5	5	3
F3				1	1	1	4	4
	P	P		P	P		P	P

Figure Q3d

(i) Identify the reference string used.

(3 marks)

(ii) If the Clock page replacement algorithm is used on the same reference string, how many page faults will be generated?

(3 marks)

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4. (a) Indicate whether the following statements about file system and I/O are true or false. Justify your answers.

(i) If a file has a symbolic link, it can be accessed using two absolute path names.

(2 marks)

(ii) Assuming the protection bits of a file are "r - x r - x - - x", only the owner of this file is able to read the file.

(2 marks)

(iii) Linked file allocation method may result in external fragmentation.

(2 marks)

(iv) If most of the files are accessed in a sequential manner, index file allocation method would be most appropriate for allocating storage space for files in such a system.

(2 marks)

(b) Consider a file system on a disk that has both logical and physical block sizes of 512 bytes. Assume that the current block of a file being accessed is logical block 6 and only the directory information about the file is in the memory. For each of the three file allocation methods (i.e., contiguous, linked, and indexed), how many physical blocks must be read from the disk if the next file block to be accessed is logical block 10?

(6 marks)

(c) A disk drive has 200 cylinders numbered from 0 to 199. At a given time, there are six outstanding requests for the following cylinders, listed in the order in which the requests arrived: 63, 102, 85, 150, 105, and 89.

(i) Assuming that the head is currently positioned over cylinder 100 and its moving direction is towards cylinder 199, calculate the total head movement required to satisfy these requests using SCAN and C-SCAN disk scheduling algorithms.

(3 marks)

(ii) Discuss the advantage of C-SCAN over SCAN disk scheduling algorithm.

(3 marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.